

Development Of TRC Plates and Angles for Compression Applications- How Thin Could we Make Them?

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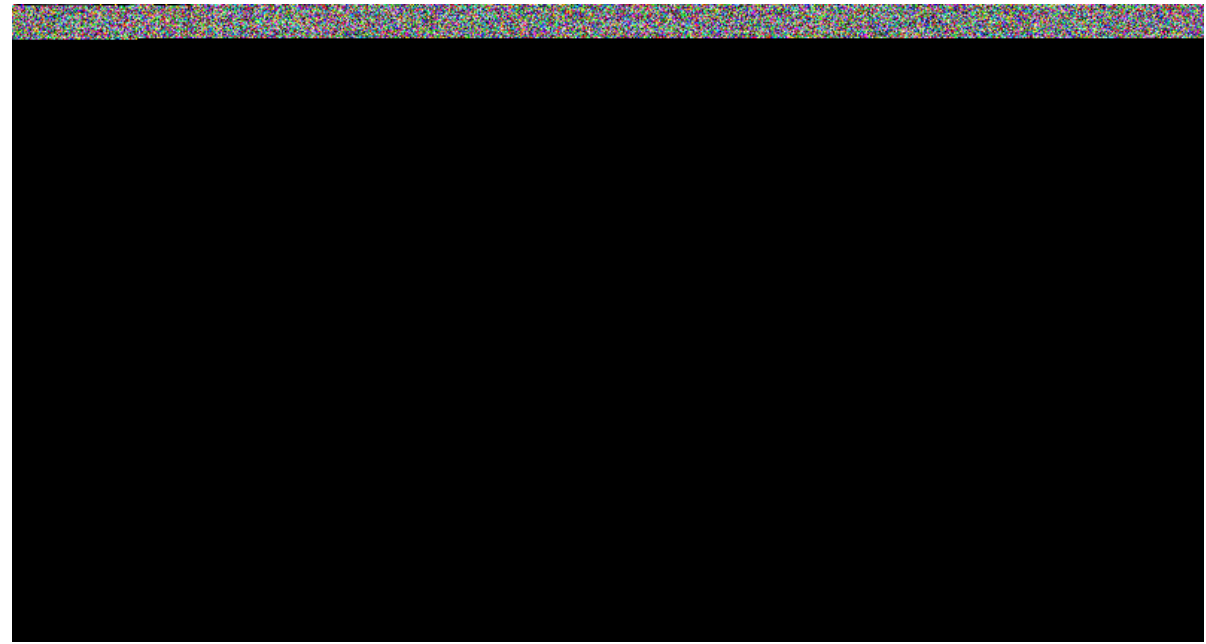
Outline

- Motivation: Sustainability, Resource Allocation, “Why not” Engineering
- Manufacturing thin section TRC structural Members
- Buckling of composite thin sections under compression and Flexure
- Finite Strip Method, FSM displacement functions
- Eigenvalue Problem, Membrane and bending actions on plates
- Signature Curve
 - Plates Theory and Parametric studies
 - ARG Plates and angles Analysis
- Conclusion

Efficiency of Ribbed Construction 1-D and 2-D



P. L. Nervi



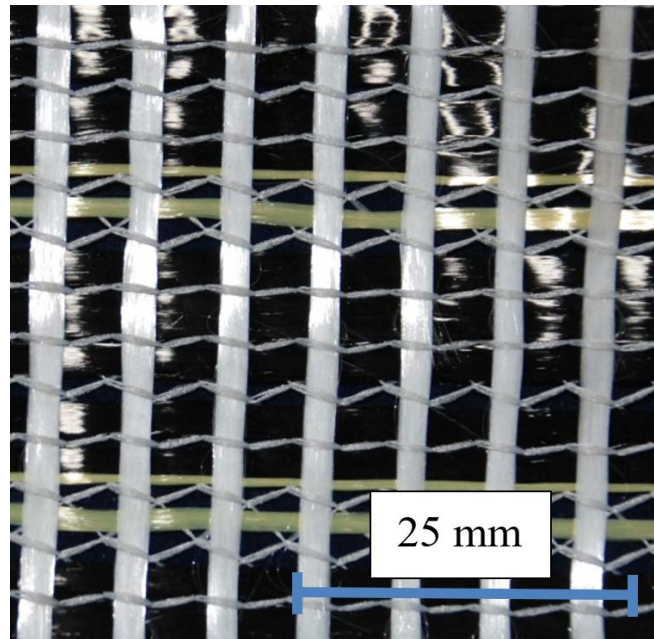
Airplane Fuselage Structural frame

Most commercial aircraft have a skin thickness between 0.07 inches (1.8 mm) to 0.09 inches (2.3 mm) depending on the location on the fuselage

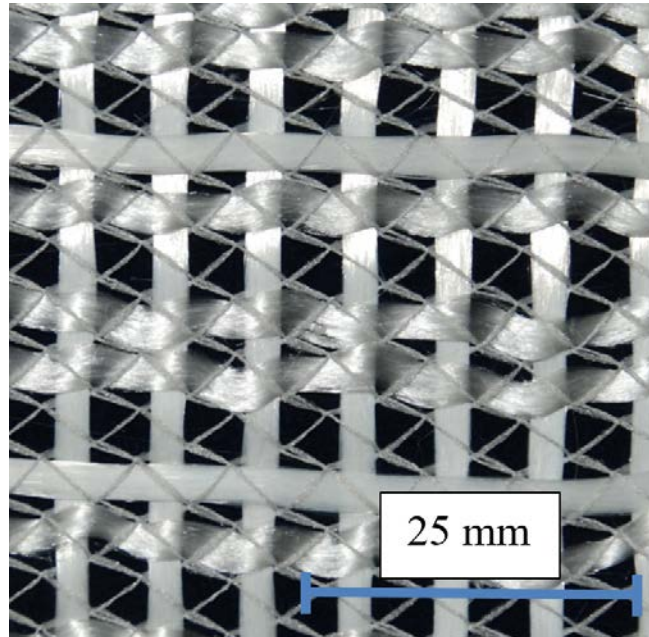
Aluminum skin in the fuselage is up to 6.5 mm, many places are as thin as 1 mm. Composite cutouts around doors are generally thicker.

TRC Structural Shapes

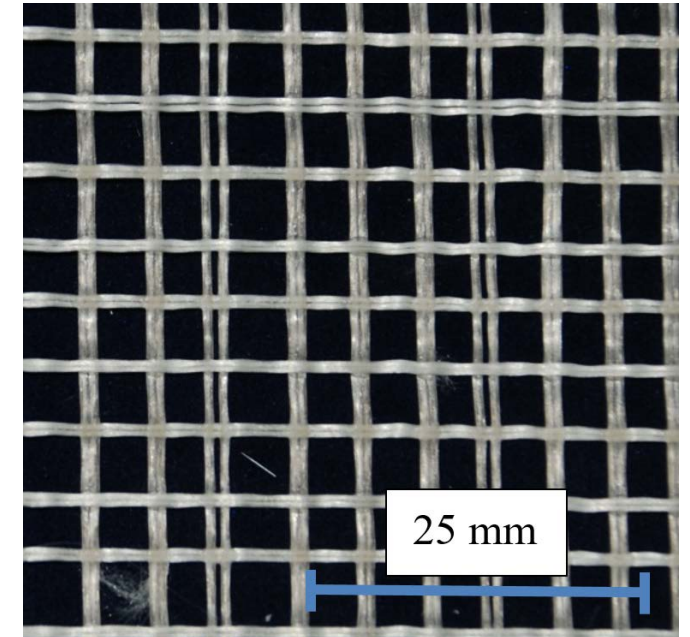
- Study properties of pultruded TRC sections
- AR Glass, Carbon, PP, and PP micro-fibers,
- Under compression
- Manufacture full-scale structural W, L and C sections
- Address the behavior under Compressive load



Carbon



PP

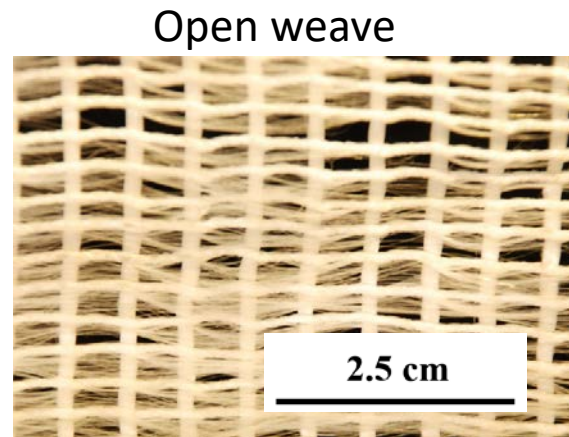


ARG

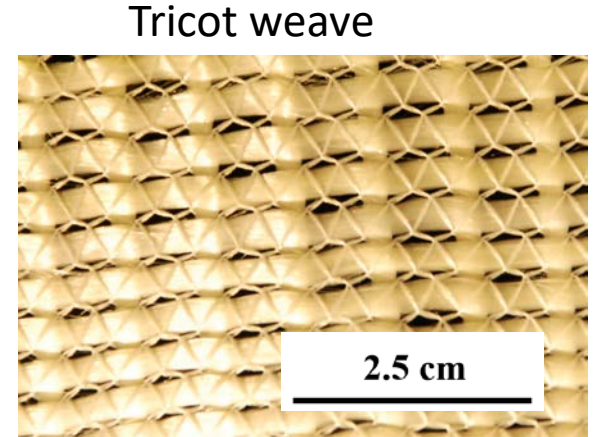
Automated Pulltrusion System



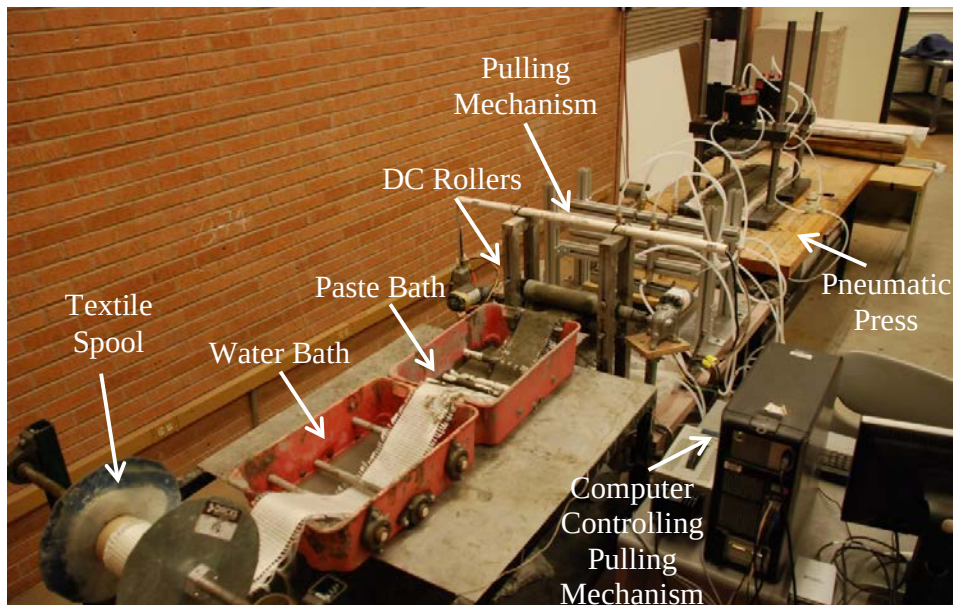
Pulling Mechanism



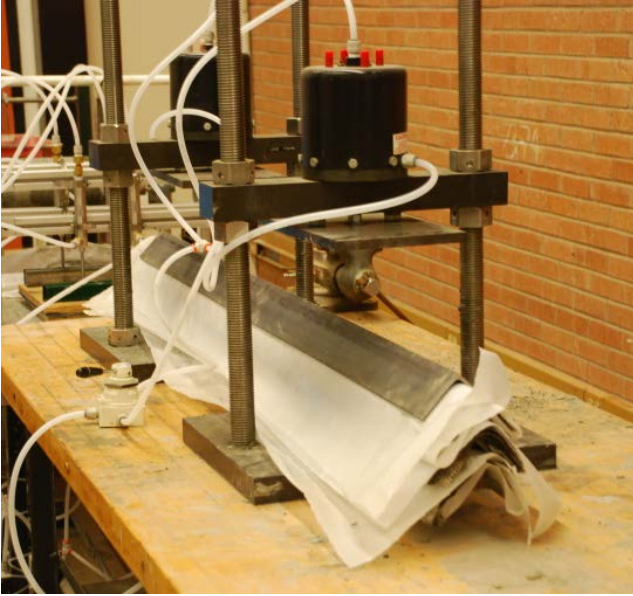
Open weave



Tricot weave



Structural Shapes – Mold Systems



Steel Angles



TRC Coupon Testing



Tension Testing Setup



Compression Testing Setup

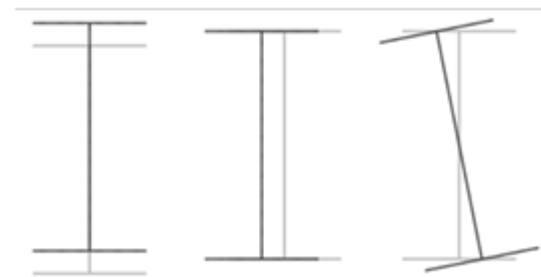
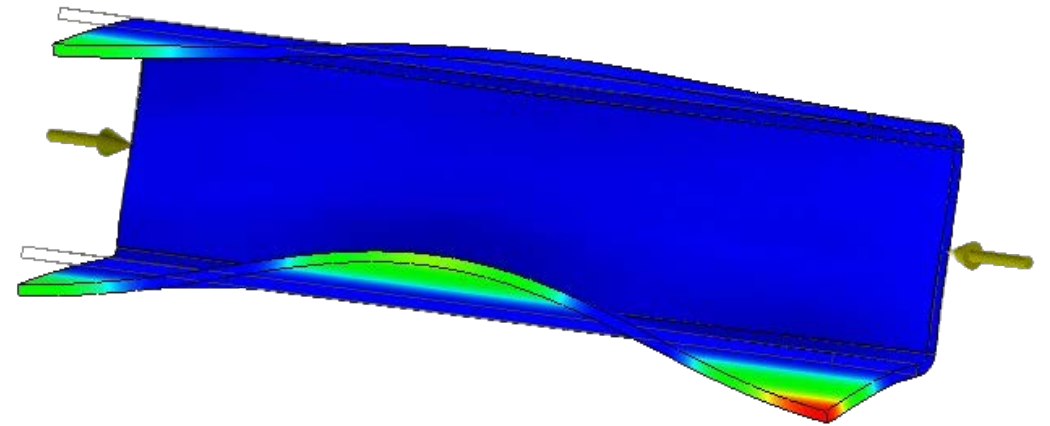
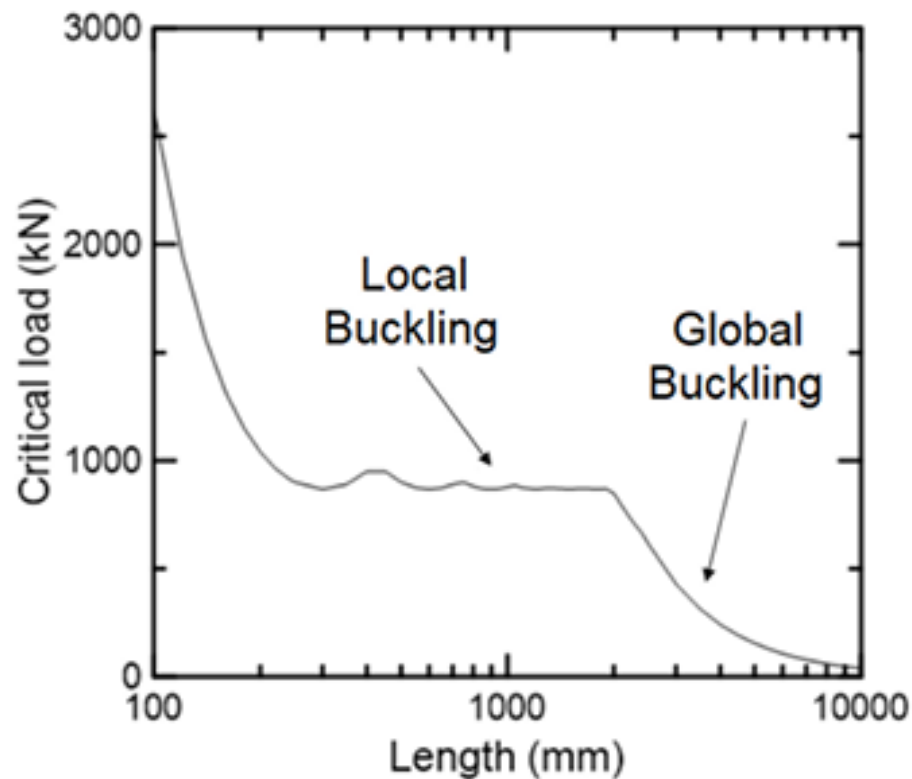
Buckling of Composite thin Sections

- Composite sections are being increasingly utilized as structural elements across various applications due to their outstanding mechanical properties relative to their weight and their superior resistance characteristics. However, despite their widespread use, there remains a significant gap in understanding regarding buckling stresses and modes for thin-walled sections.



Local Buckling- How thin can we make the stiffened and unstiffened elements?

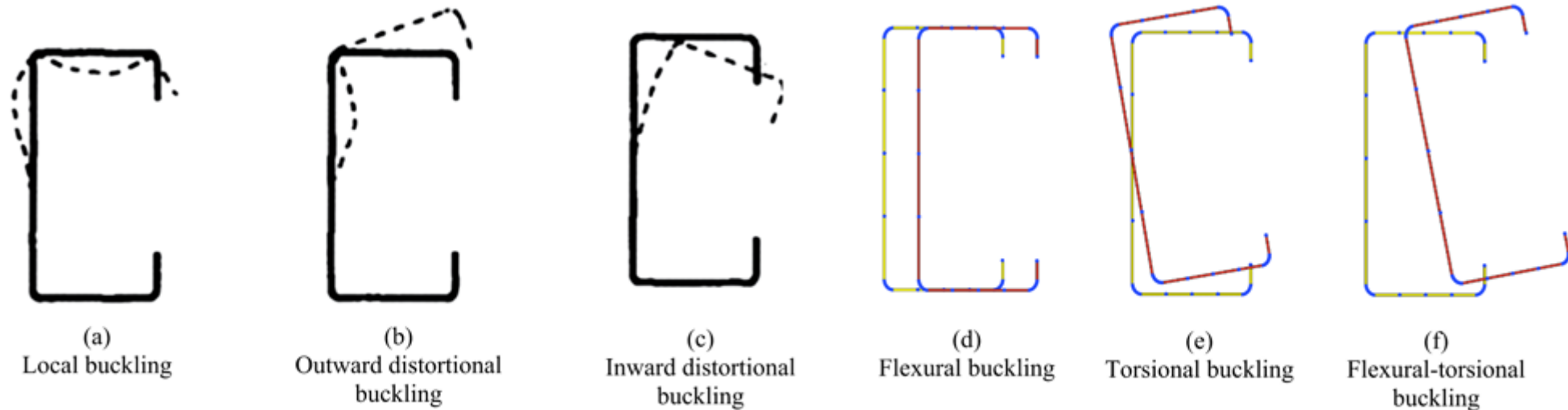
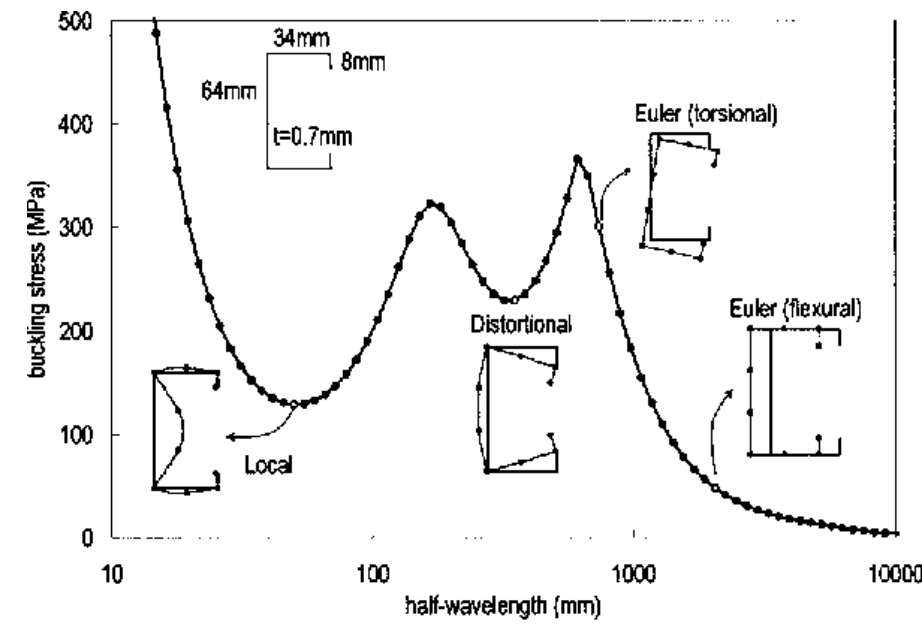
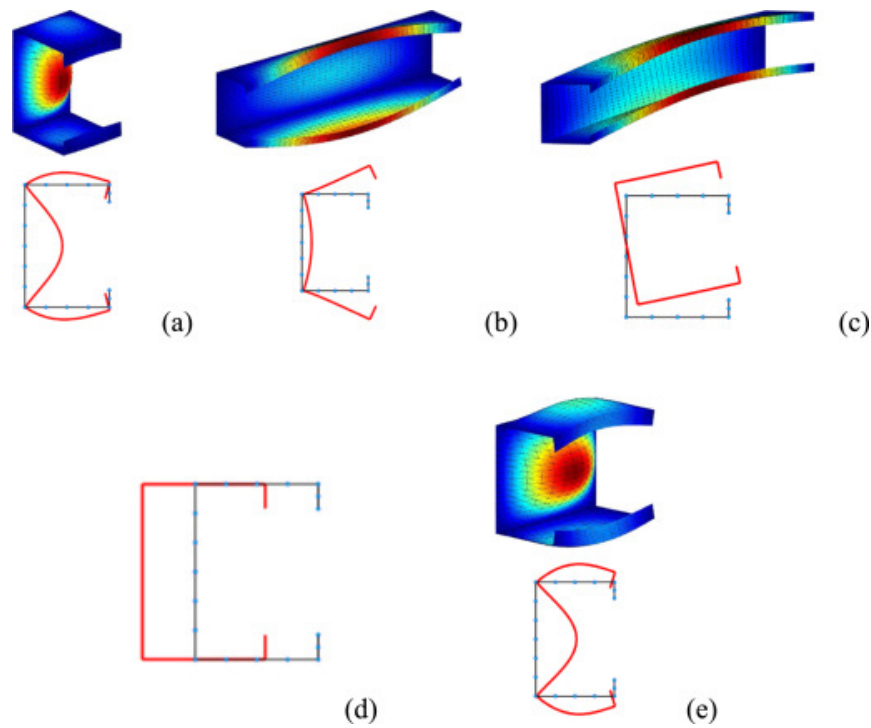
- Buckling due to compressive or flexural causes the structure to deform laterally or sideways, which can ultimately result in instability and failure.



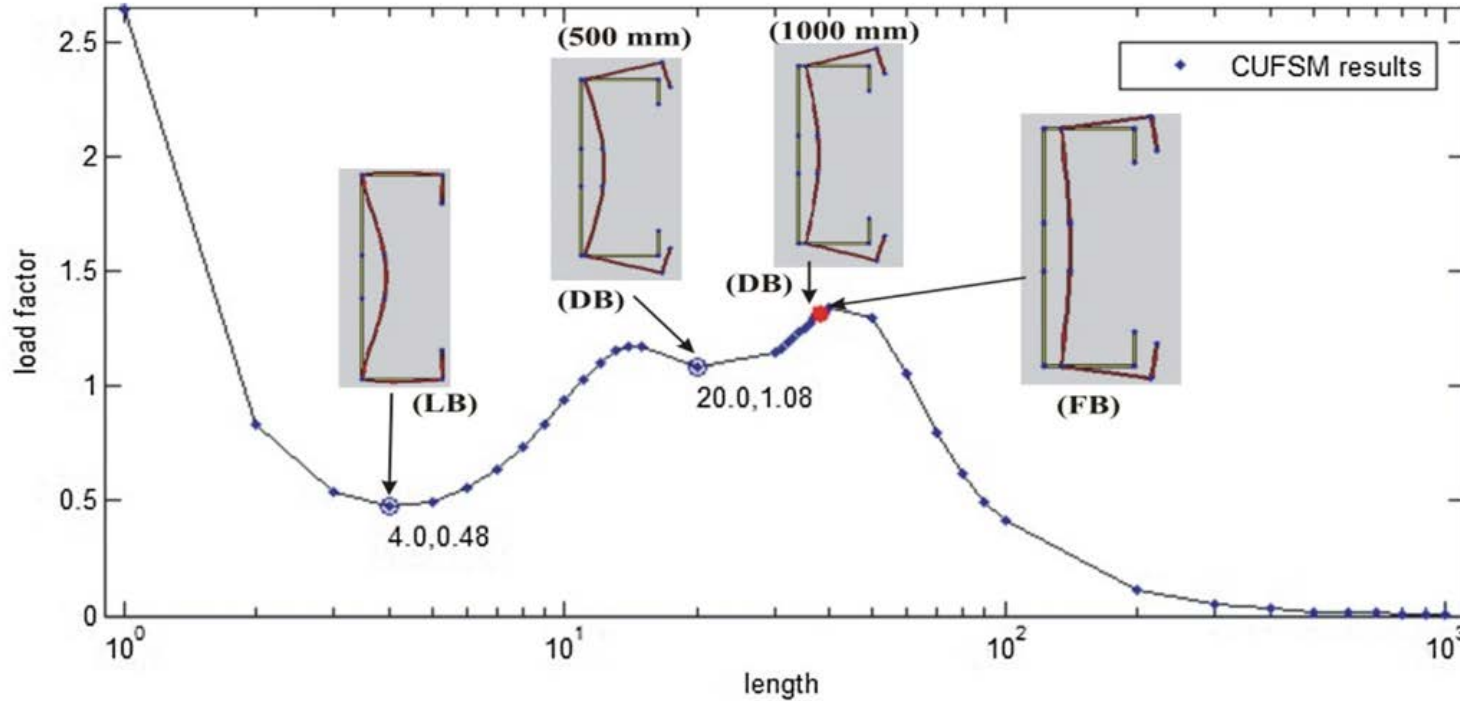
Global buckling: (a) major axis flexural buckling; (b) minor axis flexural buckling; (c) torsional buckling.



Local Buckling – Distortional Buckling - Global Buckling

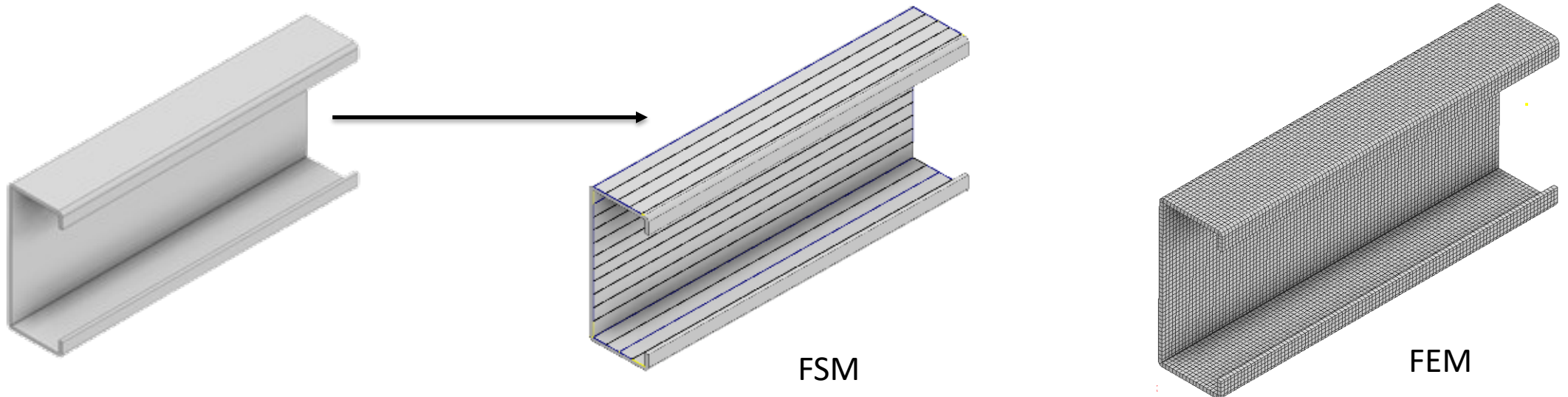


Buckling plot of Lipped Channel



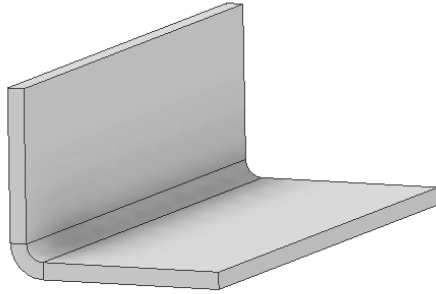
Finite Strip Method

- Pioneered by Y.K. Cheung in 1968, is an efficient tool for analyzing structures with regular geometric configurations and simple boundary conditions.
- Ability to simplify complex two-dimensional problems into one-dimensional problems
- FSM was originally developed for rectangular plate problems, similar to the classical solutions.
- Applicable to complex structures such as curved plates, skewed plates, folded plates, and box girders.
- Versatile enough to be applied to vibration and stability problems of plates and shells, making it an indispensable tool in modern structural engineering.



Input and definitions

Geometry

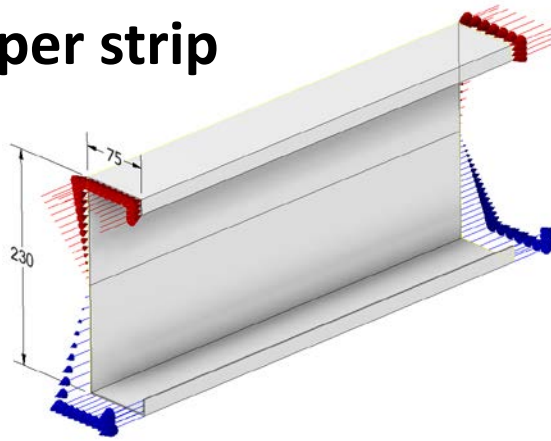


Material Properties (Orthotropic or isotropic Material)

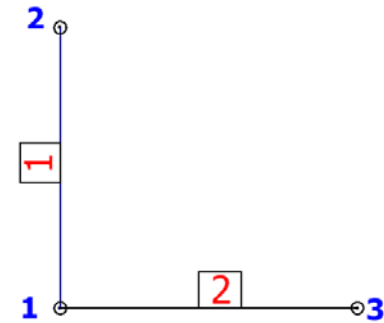
$$E_x, E_y, \nu_x, \nu_y$$

Young Modulus and Poisson's ratio

Initial Stress per strip



Assign nodes and elements



Membrane and Bending action on plates

DOFs: Three displacement and one rotation

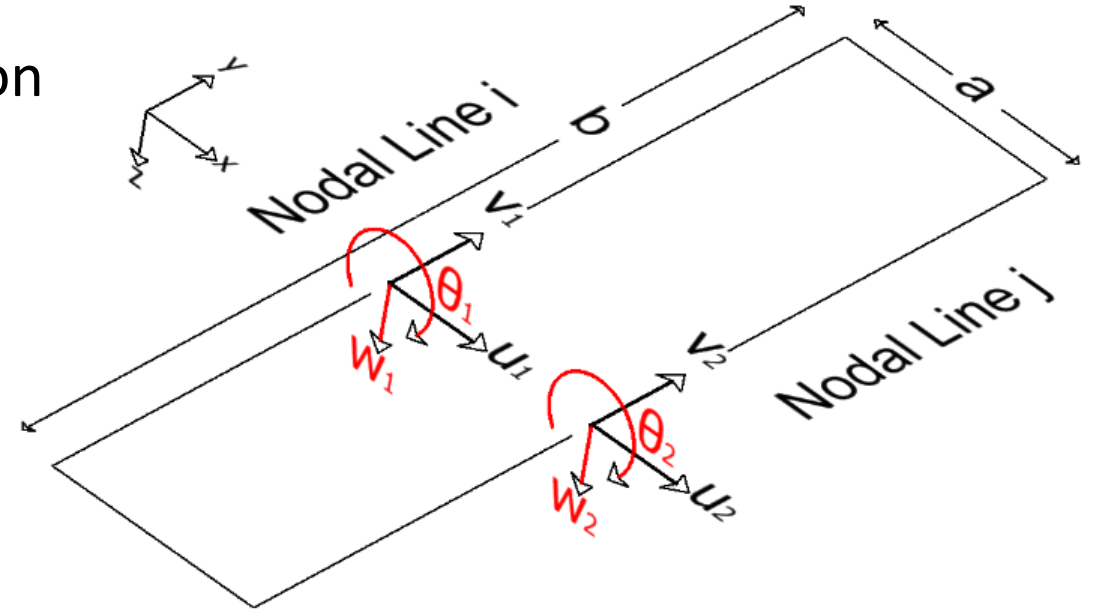


Plate Membrane: refers to a plate that has no bending rigidity, all stresses are in the plane of the surface and are usually tension

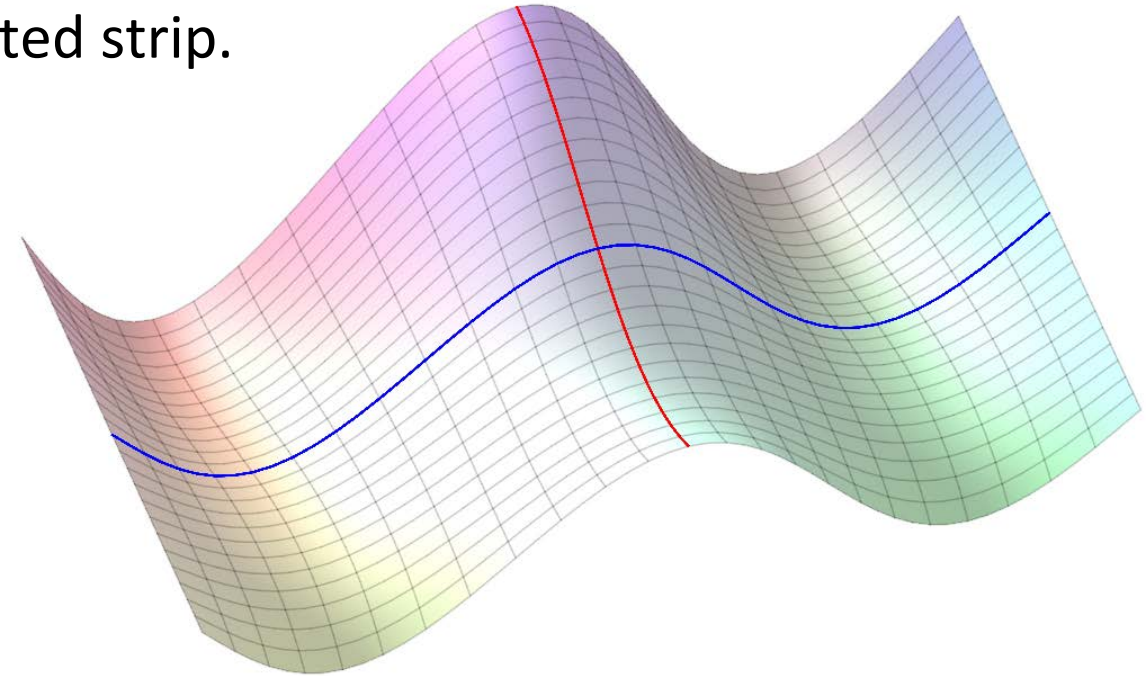
u, v

Plate Bending: refers to the deflection of a plate perpendicular to the plane of the plate under the action of external forces and moments.

w, θ

FSM Displacement Functions

Displacement function $W(x,y)$ with the Polynomial Function (red) and Sinusoidal Function (blue) in a simply supported strip.



Blue

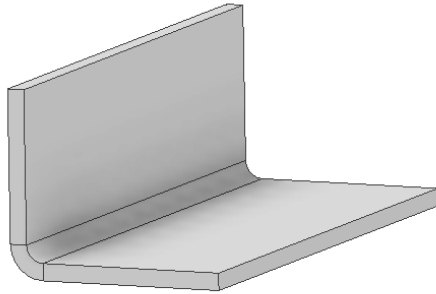
$$Y_n(y) = \sin \frac{\lambda_n y}{b} \quad \lambda_n = \pi, 2\pi, 3\pi \dots$$

Red

$$w(x) = \sum_{n=1}^r \left[\left(1 - \frac{3x^2}{a^2} + \frac{2x^3}{a^3} \right) w_{in} + \left(x - \frac{2x^2}{a} + \frac{x^3}{a^2} \right) \theta_{in} + \left(\frac{3x^2}{a^2} + \frac{2x^3}{a^3} \right) w_{jn} + \left(-\frac{x^2}{a} + \frac{x^3}{a^2} \right) \theta_{jn} \right]$$

Finite Strip Method: Input and definitions

Geometry

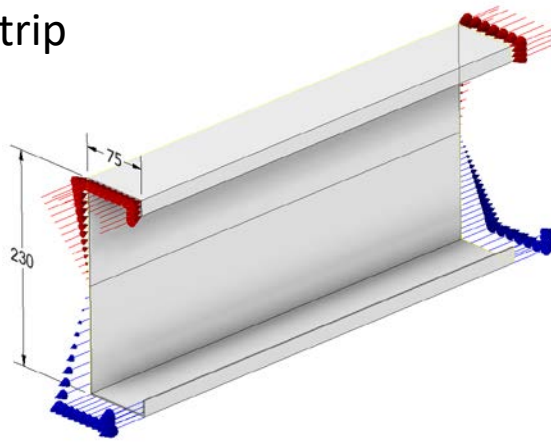


Material Properties (Orthotropic or isotropic Material)

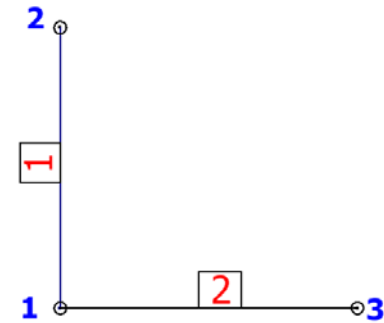
$$E_x, E_y, \nu_x, \nu_y$$

Young Modulus and Poisson's ratio

Initial Stress per strip



Assign nodes and elements



Eigenvalue Problem

$$[K][\Phi] = [\Lambda][KG][\Phi]$$

$[K]$ assembled Global stiffness matrix
 $[KG]$ assembled Global geometric matrix
 $[\Lambda]$ Eigenvalue Matrix
 $[\Phi]$ Eigenvector Matrix

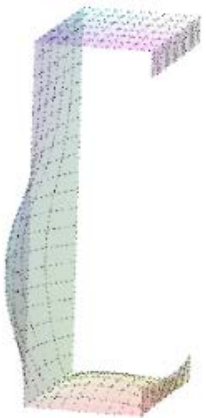
$$K = \begin{bmatrix} & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & ke_{uv} & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & ke_{w\theta} & & \\ 0 & 0 & 0 & 0 & & & & \end{bmatrix}$$

$$KG = \begin{bmatrix} & & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & kg_{uv} & & & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & & & \\ 0 & 0 & 0 & 0 & & & kg_{w\theta} & \\ 0 & 0 & 0 & 0 & & & & \end{bmatrix}$$

Signature Curve

$$[K][\Phi] = [\Lambda][KG][\Phi]$$

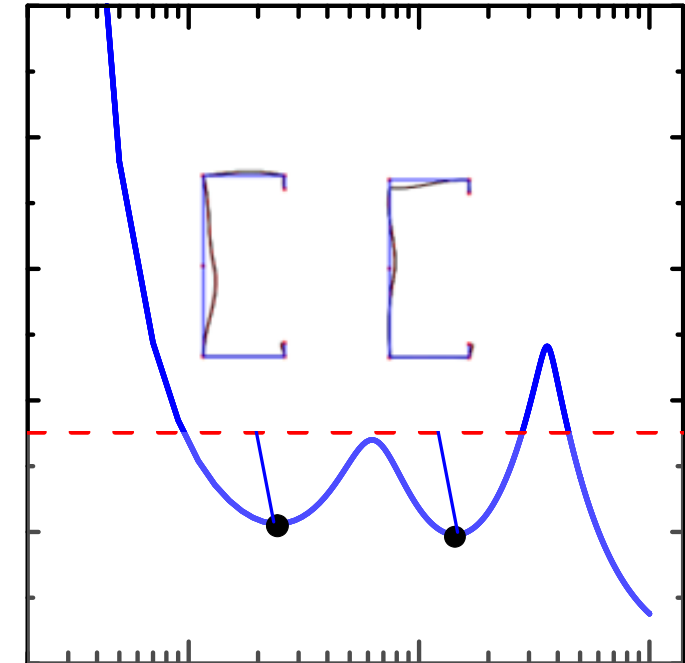
Loop over several lengths



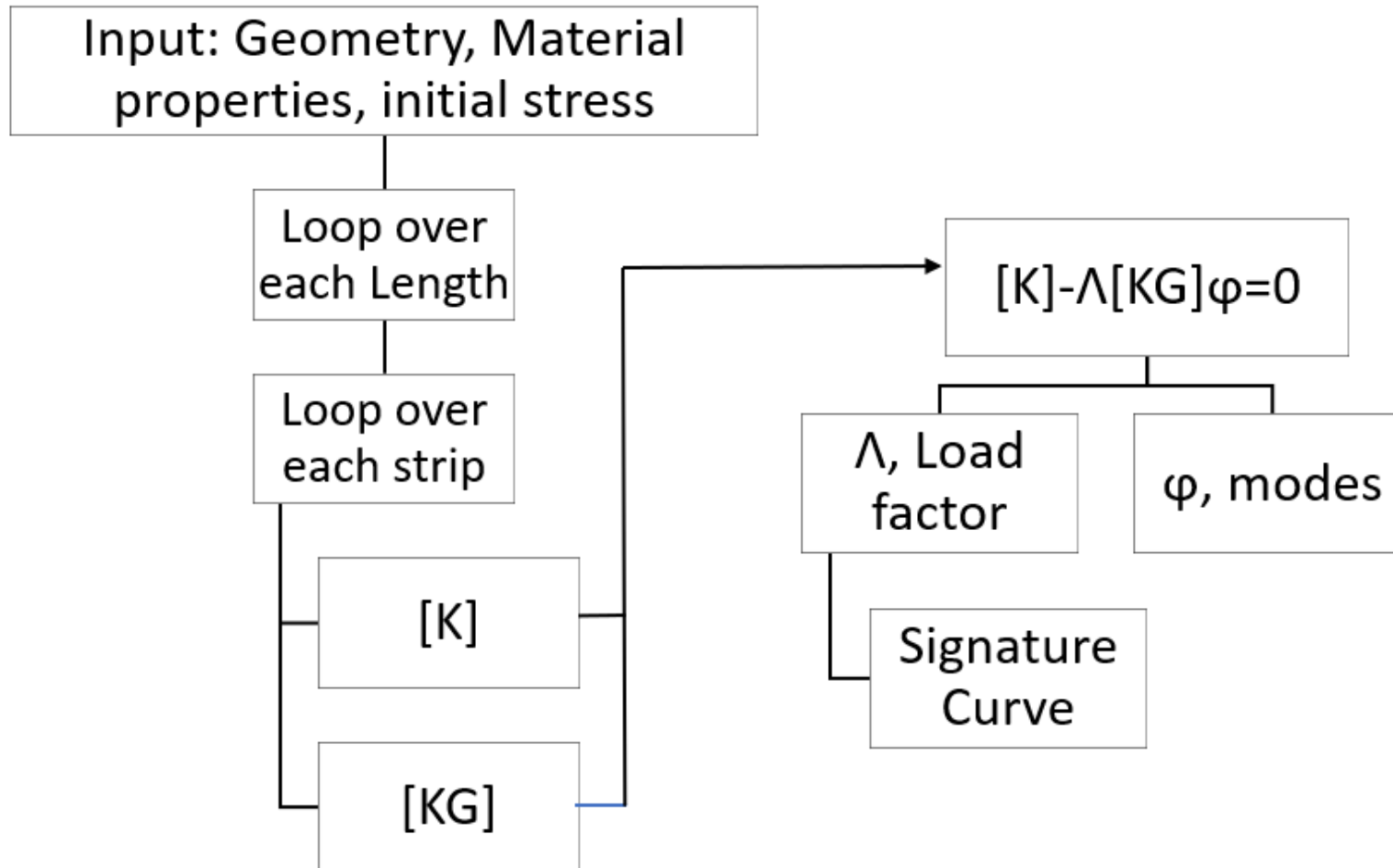
Local Buckling



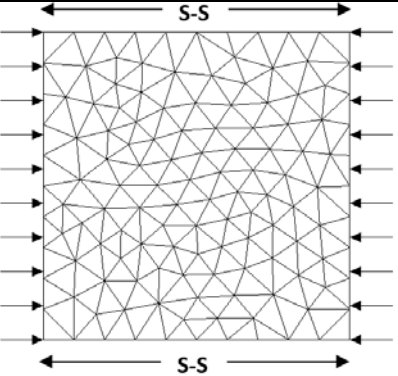
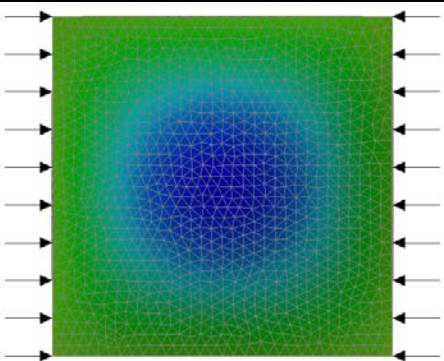
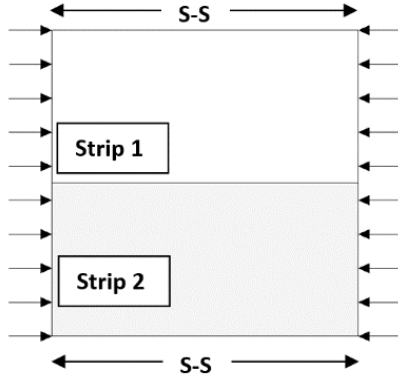
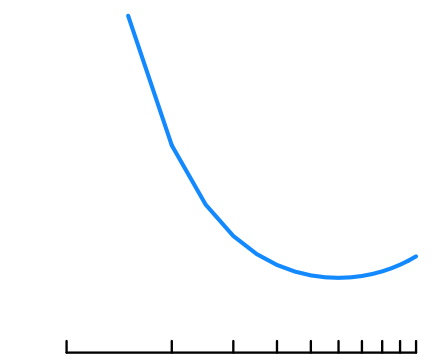
Distortional Buckling



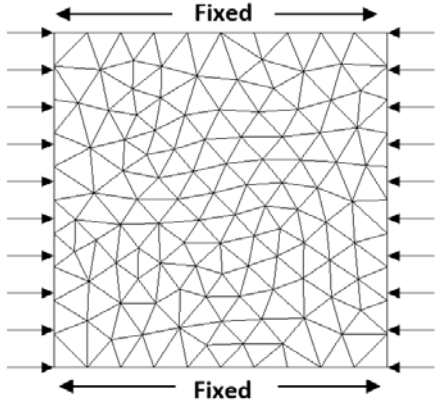
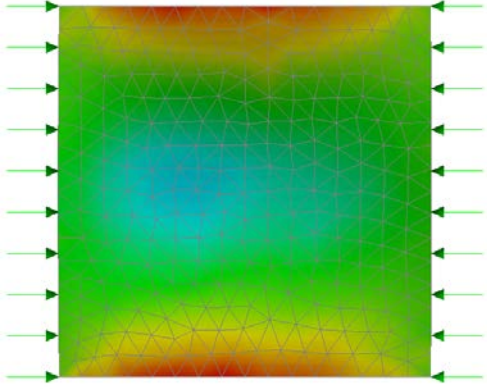
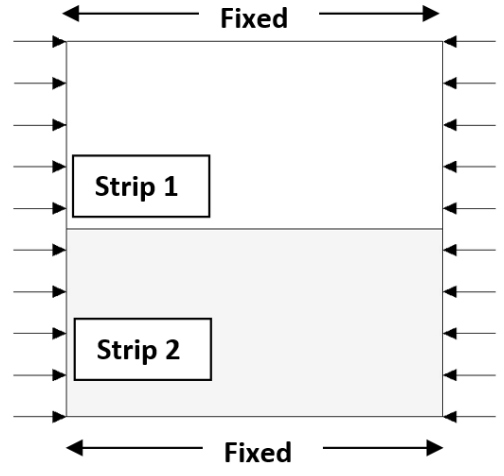
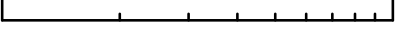
Finite Strip Method-Flow Chart



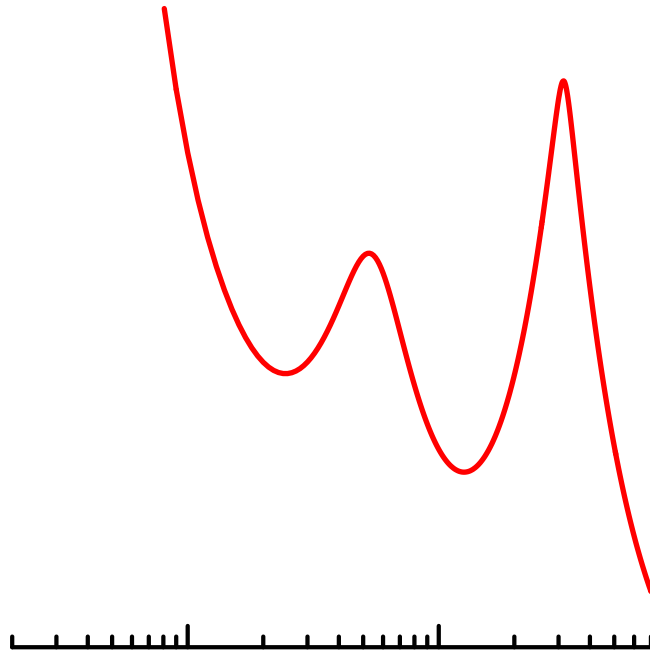
Validation Case 1: Buckling of Plates

Method	Mesh	Solution	results
FEM			$\sigma_c = 10.8 \text{ ksi}$ $k = 4.07$
FSM			$\sigma_c = 10.7 \text{ ksi},$ $k = 4.0$
SS-SS Plates	Simply-Supported plate	$\sigma_c = \frac{k\pi^2 E}{12(1-\nu^2)(b/t)^2}$	$\sigma_c = 10.7 \text{ ksi}$ $k = 4 \text{ [14]}$

Fixed-Fixed Case

Method	Mesh	Solution	results
FEM Fixed-Fixed			$L = 6in$ $\sigma_c = 23.9ksi,$ $k = 8.98$
FSM Fixed-Fixed			$\sigma_c = 22.9ksi$ $k = 8.6$
Fixed_ Fixed	Fixed-Fixed Plate $a / b = 1$	$\sigma_c = \frac{k\pi^2 E}{12(1-\nu^2)(b/t)^2}$	$\sigma_c = 21.32ksi$ $k = 8.0$ (Gerard and Becker 1957)

Parametric Studies

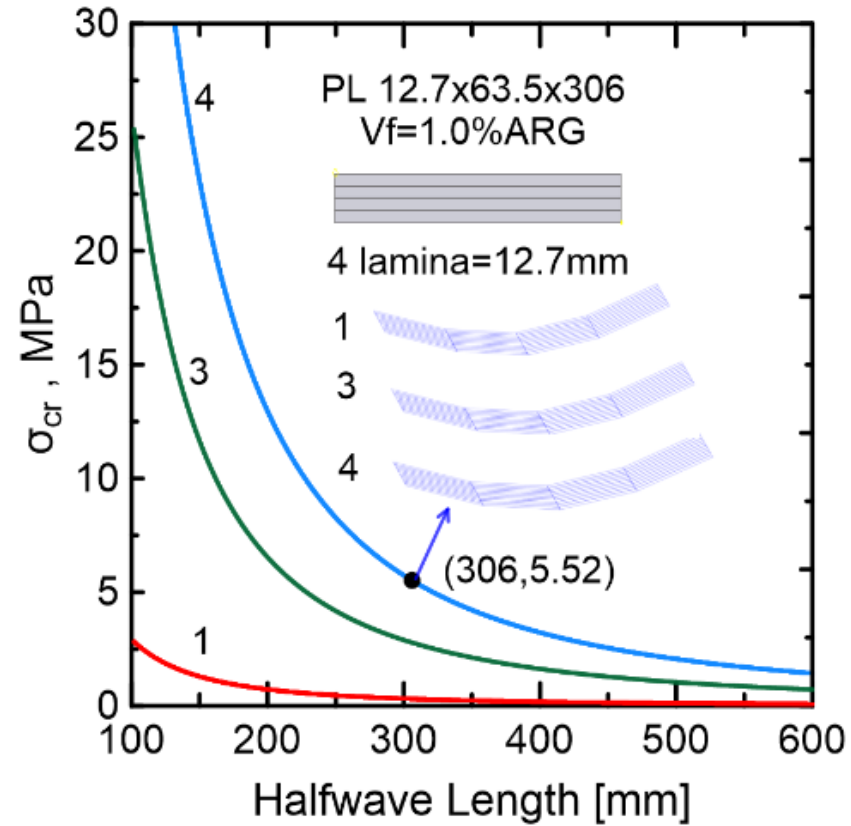


TRC Plates in compression



Compression stress-strain curve of Alkali-Resistant Glass (ARG)

TRC Plates in compression

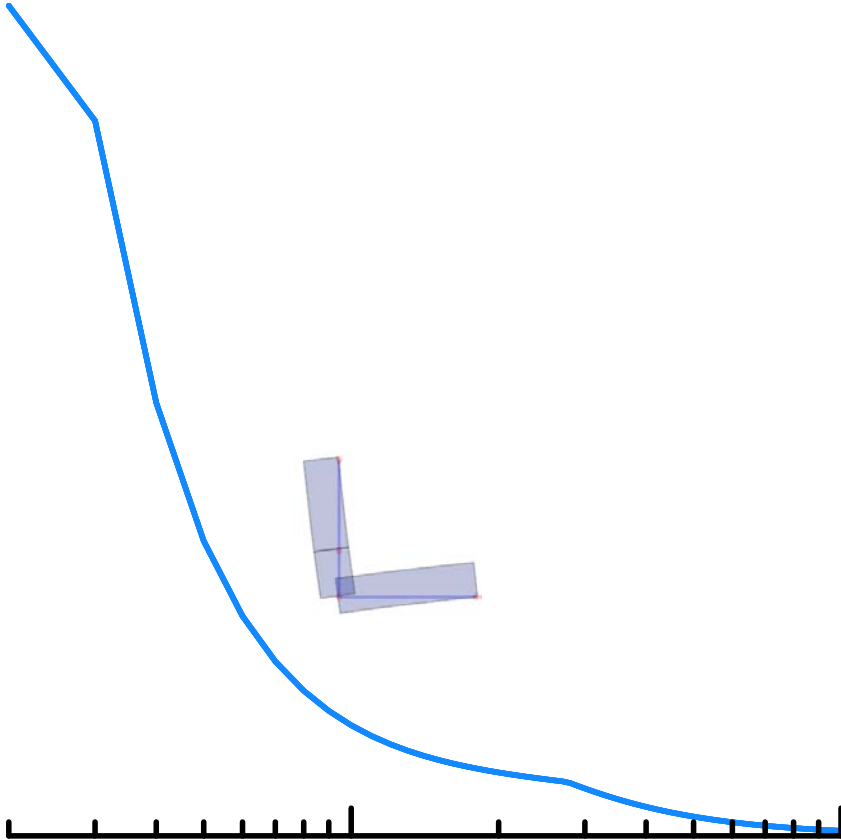


Signature curve of Alkali-Resistant Glass (ARG) Plates

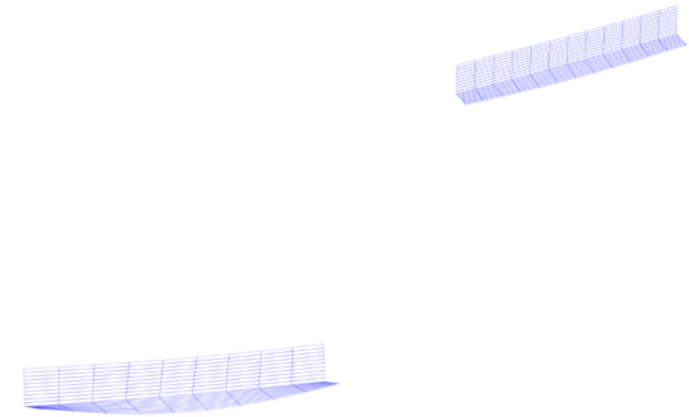
TRC L Shapes in compression



TRC L Shapes in compression



Validation Case 3: Compression testing of Glass-Epoxy L shapes



Validation Case 4: TRC I-Columns

- Experimental concentric compression tests on I-section columns with different lengths and simply supported ends.
- 10 I-TRC columns with various lengths to evaluate their stability.
- Longer columns predominantly exhibited stability failure, whereas shorter columns demonstrated failure due to the combination of axial load and bending



Conclusions

- Buckling analysis of concrete composite members was explored utilizing the Finite Strip Method.
- Development of methodology and analysis of ARG plates and L shapes were conducted
- Finite Strip Method excels in predicting buckling loads and mode shapes in concrete thin composite members.
- This methodology offers computational efficiency compared to traditional finite strip methods and significantly enhances the understanding of buckling behavior in beams with varying geometries, isotropic and orthotropic materials, and different boundary conditions.